JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Integral Calculus (2005–2026 Archive)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
66	136 M	198 Min	MCQ • MSQ • NAT (2005–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	35	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	3	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	28	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q55 • ID: JAM_2026_Q55) NAT +2 M Neg: Nil (0)

Q.55	The volume of the tetrahedron bounded by the planes $x = 1, y = 2, z = 3$ and $12x + 8y + 6z = 70$ is ______(*rounded off to one decimal place*).

Question 2 (JAM 2026 • Q53 • ID: JAM_2026_Q53) NAT +2 M Neg: Nil (0)

Q.53	The double integral of $f(x, y) = x$ over the triangular region with vertices at $\left(-\frac{1}{2}, \frac{1}{2}\right)$, $(1,2)$ and $(1, -1)$ is ______(*rounded off to one decimal place*).

Question 3 (JAM 2026 • Q22 • ID: JAM_2026_Q22) **MCQ** **+2 M** **Neg: −0.67**

Q.22	Let S be the solid of intersection of two solid spheres S_1, S_2 given by $S_1: x^2+y^2+(z-1)^2 \leq 4$ and $S_2: x^2+y^2+(z+1)^2 \leq 4$. Which ONE of the following iterated integrals expresses the volume of S?
(A)	$8\int_0^{\sqrt{3}}\int_0^{\sqrt{3-x^2}}\left(\sqrt{4-x^2-y^2}-1\right)dy\,dx$
(B)	$8\int_0^{2}\int_0^{\sqrt{4-x^2}}\left(\sqrt{4-x^2-y^2}-1\right)dy\,dx$
(C)	$2\int_0^{2}\int_0^{\sqrt{4-x^2}}\left(\sqrt{4-x^2-y^2}-1\right)dy\,dx$
(D)	$2\int_0^{\sqrt{3}}\int_0^{\sqrt{3-x^2}}\left(\sqrt{4-x^2-y^2}-1\right)dy\,dx$

Question 4 (JAM 2026 • Q20 • ID: JAM_2026_Q20) MCQ +2 M Neg: −0.67

Q.20	The semi-circle $y = \sqrt{6x - 5 - x^2}$ is rotated clockwise by the angle $\frac{\pi}{2}$ in the plane about the origin. What is the area of the planar region traced out by the semi-circle?
(A)	6π
(B)	4π
(C)	2π
(D)	π

Question 5 (JAM 2026 • Q10 • ID: JAM_2026_Q10)   MCQ   +1 M   Neg: −0.33

Q.10	Let T be the triangular region in the plane with vertices at (0,0), (1,0) and (1,1). Let f be a function defined from T to $\mathbb{R}$ given in polar coordinates. Then the double integral of f over T is equal to ______.
(A)	$\int_0^{\sqrt{2}}\left(\int_0^{\pi/4} f\, d\theta\right) r\, dr + \int_1^{\sqrt{2}}\left(\int_0^{\cos^{-1}(1/r)} f\, d\theta\right) rdr$
(B)	$\int_0^{\sqrt{2}}\left(\int_0^{\pi/4} f\, d\theta\right) rdr - \int_1^{\sqrt{2}}\left(\int_0^{\cos^{-1}(1/r)} f\, d\theta\right) rdr$
(C)	$\int_0^{\sqrt{2}}\left(\int_0^{\pi/4} f\, d\theta\right) r\, dr + \int_1^{\sqrt{2}}\left(\int_0^{\sin^{-1}(1/r)} f\, d\theta\right) rdr$
(D)	$\int_0^{\sqrt{2}}\left(\int_0^{\pi/4} f\, d\theta\right) r\, dr - \int_1^{\sqrt{2}}\left(\int_0^{\sin^{-1}(1/r)} f\, d\theta\right) rdr$

Question 6 (JAM 2026 • Q5 • ID: JAM_2026_Q5) MCQ +1 M Neg: −0.33

Q.5	Let $f:\mathbb{R}^3 \to \mathbb{R}$ be a continuous function. Then $$\int_0^2\int_0^x\int_0^y f(x,y,z)\,dz\,dy\,dx = \_\_\_\_.$$
(A)	$\int_0^2\int_0^y\int_y^2 f(x,y,z)\,dx\,dz\,dy$
(B)	$\int_0^2\int_0^x\int_z^2 f(x,y,z)\,dy\,dz\,dx$
(C)	$\int_0^2\int_y^2\int_0^x f(x,y,z)\,dz\,dx\,dy$
(D)	$\int_0^2\int_z^2\int_0^x f(x,y,z)\,dy\,dx\,dz$

Question 7 (JAM 2025 • Q57 • ID: JAM_2025_Q57) NAT +2 M Neg: Nil (0)

Q.57 Let Ω be the solid bounded by the planes $z=0$, $y=0$, $x=\frac{1}{2}$, $2y=x$ and $2x+y+z=4$.

If V is the volume of Ω, then the value of $64V$ is equal to __________ *(rounded off to two decimal places).*

Question 8 (JAM 2025 • Q48 • ID: JAM_2025_Q48) NAT +1 M Neg: Nil (0)

Q.48 Let T denote the triangle in the xy plane bounded by the x axis and the lines $y = x$ and $x = 1$. The value of the double integral (over T)

$$\iint_T (5 - y)\,dx\,dy$$

is equal to ________ *(rounded off to two decimal places).*

Question 9 (JAM 2025 • Q45 • ID: JAM_2025_Q45) NAT +1 M Neg: Nil (0)

Q.45 Let S be the surface area of the portion of the plane $z = x + y + 3$, which lies inside the cylinder $x^2 + y^2 = 1$.

Then, the value of $\left(\frac{S}{\pi}\right)^2$ is equal to ___________ *(rounded off to two decimal places).*

Question 10 (JAM 2025 • Q28 • ID: JAM_2025_Q28) MCQ +2 M Neg: −0.67

Q.28 Let Ω be the bounded region in $\mathbb{R}^3$ lying in the first octant ($x \geq 0$, $y \geq 0$, $z \geq 0$), and bounded by the surfaces $z = x^2 + y^2$, $z = 4$, $x = 0$ and $y = 0$.

Then, the volume of Ω is equal to

(A) π

(B) 2π

(C) 3π

(D) 4π

Question 11 (JAM 2025 • Q10 • ID: JAM_2025_Q10) MCQ +1 M Neg: −0.33

Q.10 The value of

$$\int_0^1 \left(\int_{\sqrt{y}}^1 3e^{x^3}\, dx \right) dy$$

is equal to

(A) $e - 1$

(B) $\frac{e-1}{2}$

(C) $\sqrt{e} - 1$

(D) $\frac{\sqrt{e}-1}{2}$

Question 12 (JAM 2024 • Q57 • ID: JAM_2024_Q57) NAT +2 M Neg: Nil (0)

Q.57 Let $c > 0$ be such that

$$\int_0^c e^{s^2}\, ds = 3$$

Then, the value of

$$\int_0^c \left(\int_x^c e^{x^2+y^2}\, dy \right) dx$$

equals ___________(*rounded off to one decimal place*).

Question 13 (JAM 2024 • Q54 • ID: JAM_2024_Q54) NAT +2 M Neg: Nil (0)

Q.54 Let

$$S = \{(x, y, z) \in \mathbb{R}^3 \colon\ x^2 + y^2 + z^2 < 1\}.$$

Then, the value of

$$\frac{1}{\pi} \iiint_S \left((x - 2y + z)^2 + (2x - y - z)^2 + (x - y + 2z)^2 \right) dxdydz$$

equals _________ (*rounded off to two decimal places*).

Question 14 (JAM 2024 • Q53 • ID: JAM_2024_Q53) NAT +2 M Neg: Nil (0)

Q.53 Let T be the planar region enclosed by the square with vertices at the points $(0,1), (1,0), (0,-1)$ and $(-1,0)$. Then, the value of

$$\iint_T \Big(\cos(\pi(x-y)) - \cos(\pi(x+y))\Big)^2 \, dx\, dy$$

equals ____________ (*rounded off to two decimal places*).

Question 15 (JAM 2024 • Q48 • ID: JAM_2024_Q48) NAT +1 M Neg: Nil (0)

Q.48 Let

$$S = \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 = 4, (x-1)^2+y^2 \leq 1, z \geq 0\}.$$

Then, the surface area of S equals ___________ (*rounded off to two decimal places*).

Question 16 (JAM 2024 • Q41 • ID: JAM_2024_Q41) NAT +1 M Neg: Nil (0)

Q.41 The area of the region

$$R = \left\{(x,y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1 \text{ and } \frac{1}{4} \leq xy \leq \frac{1}{2}\right\}$$

is __________ (*rounded off to two decimal places*).

Question 17 (JAM 2024 • Q27 • ID: JAM_2024_Q27) **MCQ** **+2 M** **Neg: −0.67**

Q.27 For $a > b > 0$, consider

$$D = \left\{(x, y, z) \in \mathbb{R}^3 \colon x^2 + y^2 + z^2 \leq a^2 \text{ and } x^2 + y^2 \geq b^2\right\}.$$

Then, the surface area of the boundary of the solid D is

(A) $4\pi(a + b)\sqrt{a^2 - b^2}$

(B) $4\pi\left(a^2 - b\sqrt{a^2 - b^2}\right)$

(C) $4\pi(a - b)\sqrt{a^2 - b^2}$

(D) $4\pi\left(a^2 + b\sqrt{a^2 - b^2}\right)$

Question 18 (JAM 2023 • Q55 • ID: JAM_2023_Q55) **NAT** **+2 M** **Neg: Nil (0)**

Q. 55 Let S be the triangular region whose vertices are $(0,0)$, $\left(0, \frac{\pi}{2}\right)$, and $\left(\frac{\pi}{2}, 0\right)$. The value of $\displaystyle\iint_S \sin(x)\cos(y)\,dx\,dy$ is equal to ____________.
(rounded off to two decimal places)

Question 19 (JAM 2023 • Q48 • ID: JAM_2023_Q48) NAT +1 M Neg: Nil (0)

Q. 48 Let V be the volume of the region $S \subseteq \mathbb{R}^3$ defined by

$$S = \{(x, y, z) \in \mathbb{R}^3 : \; xy \le z \le 4, \; 0 \le x^2 + y^2 \le 1\}.$$

Then $\dfrac{V}{\pi}$ is equal to ____________. (rounded off to two decimal places)

Question 20 (JAM 2023 • Q31 • ID: JAM_2023_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 For each $t \in (0, 1)$, the surface P_t in $\mathbb{R}^3$ is defined by

$$P_t = \{(x, y, z) : (x^2 + y^2)z = 1, \; t^2 \le x^2 + y^2 \le 1\}.$$

Let $a_t \in \mathbb{R}$ be the surface area of P_t. Then

(A) $a_t = \displaystyle\iint_{t^2 \le x^2+y^2 \le 1} \sqrt{1 + \frac{4x^2}{(x^2+y^2)^4} + \frac{4y^2}{(x^2+y^2)^4}}\, dx\, dy$

(B) $a_t = \displaystyle\iint_{t^2 \le x^2+y^2 \le 1} \sqrt{1 + \frac{4x^2}{(x^2+y^2)^2} + \frac{4y^2}{(x^2+y^2)^2}}\, dx\, dy$

(C) the limit $\displaystyle\lim_{t \to 0^+} a_t$ does NOT exist

(D) the limit $\displaystyle\lim_{t \to 0^+} a_t$ exists

Question 21 (JAM 2023 • Q26 • ID: JAM_2023_Q26) MCQ +2 M Neg: −0.67

Q. 26 Let

$$f(x,y) = \iint_{(u-x)^2+(v-y)^2\le 1} e^{-\sqrt{(u-x)^2+(v-y)^2}}\,du\,dv.$$

Then $\lim_{n\to\infty} f(n,n^2)$ is

(A) non-existent

(B) 0

(C) $\pi(1-e^{-1})$

(D) $2\pi(1-2e^{-1})$

Question 22 (JAM 2023 • Q18 • ID: JAM_2023_Q18) MCQ +2 M Neg: −0.67

Q. 18 The area of the curved surface

$$S = \left\{(x,y,z)\in\mathbb{R}^3 : z^2 = (x-1)^2+(y-2)^2\right\}$$

lying between the planes $z=2$ and $z=3$ is

(A) $4\pi\sqrt{2}$

(B) $5\pi\sqrt{2}$

(C) 9π

(D) $9\pi\sqrt{2}$

Question 23 (JAM 2023 • Q7 • ID: JAM_2023_Q7) MCQ · +1 M · Neg: −0.33

Q. 7 The value of

$$\int_0^1 \int_0^{1-x} \cos(x^3+y^2)\,dy\,dx - \int_0^1 \int_0^{1-y} \cos(x^3+y^2)\,dx\,dy$$

is

(A) 0

(B) $\frac{\cos(1)}{2}$

(C) $\frac{\sin(1)}{2}$

(D) $\cos\left(\frac{1}{2}\right) - \sin\left(\frac{1}{2}\right)$

Question 24 (JAM 2022 • Q52 • ID: JAM_2022_Q52) NAT · +2 M · Neg: Nil (0)

Q.52 If G is the region in $\mathbb{R}^2$ given by

$$G = \left\{(x,y) \in \mathbb{R}^2 : x^2+y^2 < 1,\ \frac{x}{\sqrt{3}} < y < \sqrt{3}x,\ x>0,\ y>0\right\}$$

then the value of

$$\frac{200}{\pi}\iint_G x^2\,dx\,dy$$

is equal to __________. (*Rounded off to two decimal places*)

Question 25 (JAM 2022 • Q48 • ID: JAM_2022_Q48) NAT +1 M Neg: Nil (0)

Q.48	Consider the region $$G = \{(x,y,z) \in \mathbb{R}^3 : 0 < z < x^2 - y^2,\ x^2 + y^2 < 1\}.$$ Then the volume of G is equal to ________. (*Rounded off to two decimal places*)

Question 26 (JAM 2022 • Q11 • ID: JAM_2022_Q11) MCQ +2 M Neg: −0.67

Q.11	Consider the open rectangle $G = \{(s,t) \in \mathbb{R}^2 :\ 0 < s < 1 \text{ and } 0 < t < 1\}$ and the map $T: G \to \mathbb{R}^2$ given by $$T(s,t) = \left(\frac{\pi s(1-t)}{2}, \frac{\pi(1-s)}{2}\right) \quad \text{for } (s,t) \in G.$$ Then the area of the image $T(G)$ of the map T is equal to
(A)	$\frac{\pi}{4}$
(B)	$\frac{\pi^2}{4}$
(C)	$\frac{\pi^2}{8}$
(D)	1

Question 27 (JAM 2021 • Q43 • ID: JAM_2021_Q43) NAT +1 M Neg: Nil (0)

Q. 43 Let $B = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$ and define $u(x, y, z) = \sin\left((1 - x^2 - y^2 - z^2)^2\right)$ for $(x, y, z) \in B$. Then the value of

$$\iiint_B \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) dxdydz$$

is _____.

Question 28 (JAM 2021 • Q5 • ID: JAM_2021_Q5) MCQ +1 M Neg: −0.33

Q. 5 Let p and t be positive real numbers. Let D_t be the closed disc of radius t centered at $(0, 0)$, i.e., $D_t = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq t^2\}$. Define

$$I(p, t) = \iint_{D_t} \frac{dxdy}{(p^2 + x^2 + y^2)^p}.$$

Then $\lim_{t\to\infty} I(p, t)$ is finite

(A) only if $p > 1$.

(B) only if $p = 1$.

(C) only if $p < 1$.

(D) for no value of p.

Question 29 (JAM 2020 • Q47 • ID: JAM_2020_Q47) NAT +1 M Neg: Nil (0)

Q. 47 If $\int_0^1 \int_{2y}^2 e^{x^2} dxdy = k(e^4 - 1)$, then k equals _______.

Question 30 (JAM 2020 • Q38 • ID: JAM_2020_Q38) MSQ +2 M Neg: Nil (0)

Q. 38 Let S be that part of the surface of the paraboloid $z = 16 - x^2 - y^2$ which is above the plane $z = 0$ and D be its projection on the xy-plane. Then the area of S equals

(A) $\iint_D \sqrt{1 + 4(x^2 + y^2)}\, dxdy$

(B) $\iint_D \sqrt{1 + 2(x^2 + y^2)}\, dxdy$

(C) $\int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2}\, drd\theta$

(D) $\int_0^{2\pi} \int_0^4 \sqrt{1 + 4r^2}\, rdrd\theta$

Question 31 (JAM 2020 • Q25 • ID: JAM_2020_Q25) **MCQ** **+2 M** **Neg: −0.67**

Q. 25 Let S be the part of the cone $z^2 = x^2 + y^2$ between the planes $z = 0$ and $z = 1$. Then the value of the surface integral $\iint_S (x^2 + y^2)\, dS$ is

(A) π (B) $\dfrac{\pi}{\sqrt{2}}$ (C) $\dfrac{\pi}{\sqrt{3}}$ (D) $\dfrac{\pi}{2}$

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Question 32 (JAM 2020 • Q24 • ID: JAM_2020_Q24) **MCQ** **+2 M** **Neg: −0.67**

Q. 24 The value of the triple integral $\iiint_V (x^2y + 1)\, dxdydz$, where V is the region given by $x^2 + y^2 \leq 1, 0 \leq z \leq 2$ is

(A) π (B) 2π (C) 3π (D) 4π

Question 33 (JAM 2020 • Q18 • ID: JAM_2020_Q18) **MCQ** **+2 M** **Neg: −0.67**

Q. 18 The area bounded by the curves $x^2 + y^2 = 2x$ and $x^2 + y^2 = 4x$, and the straight lines $y = x$ and $y = 0$ is

(A) $3\left(\dfrac{\pi}{2} + \dfrac{1}{4}\right)$ (B) $3\left(\dfrac{\pi}{4} + \dfrac{1}{2}\right)$ (C) $2\left(\dfrac{\pi}{4} + \dfrac{1}{3}\right)$ (D) $2\left(\dfrac{\pi}{3} + \dfrac{1}{4}\right)$

Question 34 (JAM 2020 • Q16 • ID: JAM_2020_Q16) MCQ +2 M Neg: −0.67

JAM 2020 MATHEMATICS - MA

Q. 16 Let S be the surface of the portion of the sphere with centre at the origin and radius 4, above the xy-plane. Let $\vec{F} = y\hat{i} - x\hat{j} + yx^3\hat{k}$. If $\hat{n}$ is the unit outward normal to S, then

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS$$

equals

(A) -32π (B) -16π (C) 16π (D) 32π

Question 35 (JAM 2019 • Q51 • ID: JAM_2019_Q51) NAT +2 M Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.

Q.51 The volume of the solid of revolution of the loop of the curve $y^2 = x^4(x+2)$ about the x-axis (round off to 2 decimal places) is ___________

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Question 36 (JAM 2019 • Q50 • ID: JAM_2019_Q50) NAT +1 M Neg: Nil (0)

Q.50 The volume of the solid bounded by the surfaces $x = 1 - y^2$ and $x = y^2 - 1$, and the planes $z = 0$ and $z = 2$ (round off to 2 decimal places) is _____________

Q. 51 – Q. 60 carry two marks each.

Question 37 (JAM 2019 • Q49 • ID: JAM_2019_Q49) NAT | +1 M | Neg: Nil (0)

Q.49 The value of the integral

$$\int_{-1}^{1}\int_{-1}^{1}|x+y|\,dx\,dy$$

(round off to 2 decimal places) is ___________

Question 38 (JAM 2019 • Q23 • ID: JAM_2019_Q23) MCQ | +2 M | Neg: −0.67

Q.23 The area of the part of the surface of the paraboloid $x^2+y^2+z=8$ lying inside the cylinder $x^2+y^2=4$ is

(A) $\frac{\pi}{2}(17^{3/2}-1)$ (B) $\pi(17^{3/2}-1)$ (C) $\frac{\pi}{6}(17^{3/2}-1)$ (D) $\frac{\pi}{3}(17^{3/2}-1)$

Question 39 (JAM 2019 • Q10 • ID: JAM_2019_Q10) MCQ | +1 M | Neg: −0.33

Q.10 The area of the surface generated by rotating the curve $x=y^3,\ 0\le y\le 1$, about the y-axis, is

(A) $\frac{\pi}{27}10^{3/2}$ (B) $\frac{4\pi}{3}(10^{3/2}-1)$ (C) $\frac{\pi}{27}(10^{3/2}-1)$ (D) $\frac{4\pi}{3}10^{3/2}$

Q. 11 – Q. 30 carry two marks each.

Question 40 (JAM 2019 • Q9 • ID: JAM_2019_Q9) MCQ | +1 M | Neg: −0.33

Q.9 The value of the integral

$$\int_{y=0}^{1}\int_{x=0}^{1-y^2} y\sin\left(\pi(1-x)^2\right)dx\,dy$$

is

(A) $\frac{1}{2\pi}$ (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{2}{\pi}$

Question 41 (JAM 2018 • Q58 • ID: JAM_2018_Q58) NAT +2 M Neg: Nil (0)

Q.58 The area of the parametrized surface

$S = \{((2+\cos u)\cos v,\ (2+\cos u)\sin v,\ \sin u) \in \mathbb{R}^3 \mid 0 \le u \le \frac{\pi}{2},\ 0 \le v \le \frac{\pi}{2}\}$

is ____________ (correct up to two decimal places).

Question 42 (JAM 2018 • Q56 • ID: JAM_2018_Q56) NAT +2 M Neg: Nil (0)

Q.56 The value of the integral

$$\int_0^1 \int_x^1 y^4 e^{xy^2}\, dy\, dx$$

is ______________ (correct up to three decimal places).

Question 43 (JAM 2018 • Q54 • ID: JAM_2018_Q54) NAT +2 M Neg: Nil (0)

Q.54 If the volume of the solid in $\mathbb{R}^3$ bounded by the surfaces

$$x=-1, \quad x=1, \quad y=-1, \quad y=1, \quad z=2, \quad y^2+z^2=2$$

is $\alpha - \pi$, then $\alpha =$ ________________

Question 44 (JAM 2018 • Q19 • ID: JAM_2018_Q19) MCQ +2 M Neg: −0.67

Q.19 Consider the region D in the yz plane bounded by the line $y = \frac{1}{2}$ and the curve $y^2 + z^2 = 1$, where $y \ge 0$. If the region D is revolved about the z-axis in $\mathbb{R}^3$, then the volume of the resulting solid is

(A) $\frac{\pi}{\sqrt{3}}$ (B) $\frac{2\pi}{\sqrt{3}}$ (C) $\frac{\pi\sqrt{3}}{2}$ (D) $\pi\sqrt{3}$

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Question 45 (JAM 2017 • Q56 • ID: JAM_2017_Q56) NAT +2 M Neg: Nil (0)

Q.56 For a real number x, define $\lceil x \rceil$ to be the smallest integer greater than or equal to x. Then

$$\int_0^1 \int_0^1 \int_0^1 \left(\lceil x \rceil + \lceil y \rceil + \lceil z \rceil \right) \, dx \, dy \, dz = \_\_\_\_\_\_\_\_$$

Question 46 (JAM 2017 • Q32 • ID: JAM_2017_Q32) MSQ +2 M Neg: Nil (0)

Q.32 The volume of the solid

$$\left\{ (x, y, z) \in \mathbb{R}^3 \,\middle|\, 1 \le x \le 2, \quad 0 \le y \le \frac{2}{x}, \quad 0 \le z \le x \right\}$$

is expressible as

(A) $\int_1^2 \int_0^{2/x} \int_0^x dz\, dy\, dx$

(B) $\int_1^2 \int_0^x \int_0^{2/x} dy\, dz\, dx$

(C) $\int_0^2 \int_1^z \int_0^{2/x} dy\, dx\, dz$

(D) $\int_0^2 \int_{\max\{z,1\}}^2 \int_0^{2/x} dy\, dx\, dz$

Question 47 (JAM 2017 • Q16 • ID: JAM_2017_Q16) MCQ +2 M Neg: −0.67

Q.16 The area of the surface $z = \frac{xy}{3}$ intercepted by the cylinder $x^2 + y^2 \le 16$ lies in the interval

(A) $(20\pi, 22\pi]$ (B) $(22\pi, 24\pi]$ (C) $(24\pi, 26\pi]$ (D) $(26\pi, 28\pi]$

Question 48 (JAM 2017 • Q6 • ID: JAM_2017_Q6) MCQ +1 M Neg: −0.33

Q.6 $$\int_0^1 \int_x^1 \sin(y^2) dy\, dx =$$

(A) $\frac{1+\cos 1}{2}$ (B) $1 - \cos 1$ (C) $1 + \cos 1$ (D) $\frac{1-\cos 1}{2}$

Question 49 (JAM 2016 • Q57 • ID: JAM_2016_Q57) NAT +2 M Neg: Nil (0)

Q.57 The value of the double integral

$$\int_0^{\pi}\int_0^{x} \frac{\sin y}{\pi - y}\, dy\, dx$$

is ____________

Question 50 (JAM 2016 • Q52 • ID: JAM_2016_Q52) NAT +2 M Neg: Nil (0)

Q.52 Let R be the region enclosed by $x^2 + 4y^2 \geq 1$ and $x^2 + y^2 \leq 1$. Then the value of

$$\iint_R |xy|\, dx\, dy \quad \text{is } \_\_\_\_\_\_\_\_$$

Question 51 (JAM 2016 • Q16 • ID: JAM_2016_Q16) MCQ +2 M Neg: −0.67

Q.16 The surface area of the portion of the plane $y + 2z = 2$ within the cylinder $x^2 + y^2 = 3$ is

(A) $\frac{3\sqrt{5}}{2}\pi$ (B) $\frac{5\sqrt{5}}{2}\pi$ (C) $\frac{7\sqrt{5}}{2}\pi$ (D) $\frac{9\sqrt{5}}{2}\pi$

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Question 52 (JAM 2016 • Q15 • ID: JAM_2016_Q15)   MCQ   +2 M   Neg: −0.67

Q.15 If the triple integral over the region bounded by the planes

$$2x + y + z = 4, \qquad x = 0, \qquad y = 0, \qquad z = 0$$

is given by

$$\int_0^2 \int_0^{\lambda(x)} \int_0^{\mu(x,y)} dz\, dy\, dx\,,$$

then the function $\lambda(x) - \mu(x, y)$ is

(A) $x + y$ (B) $x - y$ (C) x (D) y

Question 53 (JAM 2015 • Q58 • ID: JAM_2015_Q58)   NAT   +2 M   Neg: Nil (0)

Q.18

Let ℓ be the length of the portion of the curve $x = x(y)$ between the lines $y = 1$ and $y = 3$, where $x(y)$ satisfies

$$\frac{dx}{dy} = \frac{\sqrt{1 + y^2 + y^4}}{y}\,, \qquad x(1) = 0$$

The value of ℓ, correct upto three decimal places, is ______________

Question 54 (JAM 2015 • Q55 • ID: JAM_2015_Q55)   NAT   +2 M   Neg: Nil (0)

Q.15

Let $f\colon \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \dfrac{y}{\sin y}, & y \neq 0 \\ 1, & y = 0 \end{cases}$$

Then the integral

$$\frac{1}{\pi^2} \int_{x=0}^{1} \int_{y=\sin^{-1} x}^{\frac{\pi}{2}} f(x, y)\, dy\, dx$$

correct upto three decimal places, is ______________

Question 55 (JAM 2015 • Q42 • ID: JAM_2015_Q42) NAT +1 M Neg: Nil (0)

Q.2

Let S be the portion of the surface $z = \sqrt{16 - x^2}$ bounded by the planes $x = 0,\ x = 2,\ y = 0$, and $y = 3$. The surface area of S, correct upto three decimal places, is ____________

Question 56 (JAM 2015 • Q26 • ID: JAM_2015_Q26) MCQ +2 M Neg: −0.67

Q.26 The area of the planar region bounded by the curves $x = 6y^2 - 2$ and $x = 2y^2$ is

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{2\sqrt{2}}{3}$ (C) $\frac{4\sqrt{2}}{3}$ (D) $\sqrt{2}$

Question 57 (JAM 2015 • Q14 • ID: JAM_2015_Q14) MCQ +2 M Neg: −0.67

Q.14 Let S be the bounded surface of the cylinder $x^2 + y^2 = 1$ cut by the planes $z = 0$ and $z = 1 + x$. Then the value of the surface integral $\iint_S 3z^2 d\sigma$ is equal to

(A) $\int_0^{2\pi} (1 + \cos\theta)^3 d\theta$

(B) $\int_0^{2\pi} \sin\theta\cos\theta\,(1 + \cos\theta)^2 d\theta$

(C) $\int_0^{2\pi} (1 + 2\cos\theta)^3 d\theta$

(D) $\int_0^{2\pi} \sin\theta\cos\theta\ (1 + 2\cos\theta)^2 d\theta$

Question 58 (JAM 2015 • Q6 • ID: JAM_2015_Q6) MCQ +1 M Neg: −0.33

Q.6 The volume of the portion of the solid cylinder $x^2 + y^2 \leq 2$ bounded above by the surface $z = x^2 + y^2$ and bounded below by the xy-plane is

(A) π (B) 2π (C) 3π (D) 4π

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Question 59 (JAM 2014 • Q27 • ID: JAM_2014_Q27) MCQ +2 M Neg: −0.67

Q.27 The value of $\int_{x=0}^{1}\int_{y=0}^{x^2}\int_{z=0}^{y}(y+2z)\,dz\,dy\,dx$ is

(A) $\frac{1}{53}$ (B) $\frac{2}{21}$ (C) $\frac{1}{6}$ (D) $\frac{5}{3}$

Question 60 (JAM 2013 • Q6 • ID: JAM_2013_Q6) MCQ +2 M Neg: −0.67

The value of the integral

$$\iint_D \sqrt{x^2+y^2}\,dx\,dy\,, \quad D=\left\{(x,y)\in\mathbb{R}^2:\ x\le x^2+y^2\le 2x\right\}$$

is

(A) 0 (B) $\frac{7}{9}$ (C) $\frac{14}{9}$ (D) $\frac{28}{9}$

Question 61 (JAM 2012 • Q4 • ID: JAM_2012_Q4) MCQ +6 M Neg: −2.0

Q.4 The value of $\int_{z=0}^{1}\int_{y=0}^{z}\int_{x=0}^{y} x\,y^2 z^3\,dx\,dy\,dz$ is

(A) $\frac{1}{90}$ (B) $\frac{1}{50}$ (C) $\frac{1}{45}$ (D) $\frac{1}{10}$

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Question 62 (JAM 2011 • Q4 • ID: JAM_2011_Q4) MCQ +6 M Neg: −2.0

Q.4 Let V be the region bounded by the planes $x=0$, $x=2$, $y=0$, $z=0$ and $y+z=1$. Then the value of the integral $\iiint_V y\,dx\,dy\,dz$ is

(A) $\frac{1}{2}$ (B) $\frac{4}{3}$ (C) 1 (D) $\frac{1}{3}$

Question 63 (JAM 2010 • Q2 • ID: JAM_2010_Q2) MCQ +6 M Neg: −2.0

Q.2 The value of $\iint_G \frac{\ln(x^2+y^2)}{x^2+y^2}\,dx\,dy$, where $G=\{(x,y)\in\mathbf{R}^2 : 1\le x^2+y^2\le e^2\}$, is

(A) π (B) 2π (C) 3π (D) 4π

Question 64 (JAM 2009 • Q8 • ID: JAM_2009_Q8) MCQ +6 M Neg: −2.0

Q.8 Let $F:\mathrm{R}\to\mathrm{R}$ be a continuous function and $a>0$. Then the integral $\int_0^a\left[\int_0^x F(y)\,dy\right]dx$ equals

(A) $\int_0^a yF(y)\,dy$ (B) $\int_0^a (a-y)\,F(y)\,dy$

(C) $\int_0^a (y-a)\,F(y)\,dy$ (D) $\int_a^0 y\,F(y)\,dy$

Question 65 (JAM 2005 • Q12 • ID: JAM_2005_Q12) MCQ +6 M Neg: −2.0

12. Let $y = f(x)$ be a smooth curve such that $0 < f(x) < K$ for all $x \in [a,b]$. Let

L = length of the curve between $x = a$ and $x = b$

A = area bounded by the curve, x-axis, and the lines $x = a$ and $x = b$

S = area of the surface generated by revolving the curve about x-axis between $x = a$ and $x = b$

Then

(A) $2\pi KL < S < 2\pi A$

(B) $S \le 2\pi A < 2\pi KL$

(C) $2\pi A \le S < 2\pi KL$

(D) $2\pi A < 2\pi KL < S$

Question 66 (JAM 2005 • Q10 • ID: JAM_2005_Q10) MCQ +6 M Neg: −2.0

10. The integral $\int_0^1 \left[\int_0^{1-z}\left(\int_0^2 dx\right)dy\right]dz$ is equal to

(A) $\int_0^1 \left[\int_0^{1-y}\left(\int_0^2 dx\right)dz\right]dy$

(B) $\int_0^1 \left[\int_0^{1-y}\left(\int_0^1 dx\right)dz\right]dy$

(C) $\int_0^2 \left[\int_0^2\left(\int_0^{1-z} dx\right)dz\right]dy$

(D) $\int_0^2 \left[\int_0^2\left(\int_0^{1-y} dx\right)dz\right]dy$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Integral Calculus (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	**4.0 to 4.0**
2	NAT	+2	**1.1 to 1.3**
3	MCQ	+2	**A**
4	MCQ	+2	**A**
5	MCQ	+1	**MTA**
6	MCQ	+1	**A**
7	NAT	+2	**MTA**
8	NAT	+1	**2.31 to 2.35**
9	NAT	+1	**2.98 to 3.02**
10	MCQ	+2	**B**
11	MCQ	+1	**A**
12	NAT	+2	**4.4 to 4.6**
13	NAT	+2	**4.70 to 4.90**
14	NAT	+2	**1.95 to 2.05**
15	NAT	+1	**4.50 to 4.60**
16	NAT	+1	**0.24 to 0.26**
17	MCQ	+2	**A**
18	NAT	+2	**0.49 to 0.51**
19	NAT	+1	**3.99 to 4.01**
20	MSQ	+2	**A, C**
21	MCQ	+2	**D**
22	MCQ	+2	**B**
23	MCQ	+1	**A**
24	NAT	+2	**4.15 to 4.17**
25	NAT	+1	**0.49 to 0.51**
26	MCQ	+2	**C**
27	NAT	+1	**0**
28	MCQ	+1	**A**
29	NAT	+1	**0.25 to 0.25**
30	MSQ	+2	**A;D**
31	MCQ	+2	**B**
32	MCQ	+2	**B**
33	MCQ	+2	**B**
34	MCQ	+2	**A**
35	NAT	+2	**6.60 to 6.80**
36	NAT	+1	**5.30 to 5.50**
37	NAT	+1	**2.60 to 2.70**
38	MCQ	+2	**C**
39	MCQ	+1	**C**
40	MCQ	+1	**A**
41	NAT	+2	**6.30 to 6.70**
42	NAT	+2	**0.230 to 0.250**
43	NAT	+2	**5.99 to 6.01**
44	MCQ	+2	**C**
45	NAT	+2	**2.9 to 3.1**
46	MSQ	+2	**A, B, D**
47	MCQ	+2	**A**
48	MCQ	+1	**D**
49	NAT	+2	**2.0 to 2.0**
50	NAT	+2	**0.35 to 0.4**
51	MCQ	+2	**A**
52	MCQ	+2	**D**
53	NAT	+2	**5.09 to 5.10**
54	NAT	+2	**0.125**
55	NAT	+1	**6.28 to 6.29**
56	MCQ	+2	**C**
57	MCQ	+2	**A**
58	MCQ	+1	**B**
59	MCQ	+2	**B**
60	MCQ	+2	**D**
61	MCQ	+6	**A**
62	MCQ	+6	**D**
63	MCQ	+6	**B**
64	MCQ	+6	**B**
65	MCQ	+6	**C**
66	MCQ	+6	**A**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.